

SAMPATH REDDY KAMSANI

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PROFESSIONAL SUMMARY

Master's student in Software Design with Artificial Intelligence at Technological University of the Shannon (TUS), Athlone, with a B.Tech in Computer Science and Engineering (AI & ML) and hands-on experience through internships and academic projects. Experienced in developing end-to-end machine learning and deep learning applications, multimodal AI systems, fraud detection solutions, and intelligent software applications using Python, TensorFlow, PyTorch, Scikit-learn, and Flask. Strong foundation in software development, object-oriented programming, REST APIs, data analysis, and problem-solving, with a keen interest in building reliable and scalable AI-driven software solutions.

EDUCATION

MSc in Software Design with Artificial Intelligence

Technological University of the Shannon (TUS), Athlone | Sep 2026 – July 2027

B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning) | GPA: 7.7/10

DRK Institute of Science & Technology, Hyderabad | Oct 2022 – Apr 2026

EXPERIENCE

Quant ML Intern — Blutick Solutions Pvt Ltd | Apr 2026– July 2026

- Learned quantitative trading concepts, machine learning techniques, and Python-based development for financial markets.
- Built Python applications for processing historical and real-time market data under mentorship.
- Applied machine learning techniques for trend forecasting, market regime detection, and portfolio analysis using historical financial datasets.
- Developed feature engineering pipelines, implemented technical indicators, and performed backtesting of quantitative trading strategies.
- Designed and implemented a Previous 3-Day High/Low Breach breakout strategy with technical indicators and backtesting for quantitative trading analysis.
- Gained practical experience in algorithmic trading workflows, data analysis, and ML model development.

Student Web Developer — DRK Institute of Science and Technology | Mar 2025– May 2026

- Designed, developed, and deployed a responsive institutional website using HTML, CSS, JavaScript, PHP, and MySQL, modernizing the college's previous website and ensuring seamless functionality across desktop and mobile devices.
- Developed a secure attendance management system with role-based access control for administrators, faculty, and students.
- Integrated an AI-powered chatbot to automate student and visitor queries related to admissions, courses, academics, and campus information.
- Enhanced website performance, navigation, accessibility, and content organization to improve the overall user experience for students, faculty, prospective applicants, and visitors.
- Enabled centralized digital access to institutional information, including departments, academic programs, admissions, placements, events, notices, and academic resources.
- Delivered a production-ready website that is currently deployed and used as the institute's official website.

SKILLS

- Programming Languages: Python, Java, C
- Machine Learning: Scikit-learn, TensorFlow, PyTorch, CNN, LSTM
- Software Development: Object-Oriented Programming (OOP), Data Structures & Algorithms, REST APIs
- NLP & Computer Vision: BERT, NLTK, OpenCV
- Data Analysis: NumPy, Pandas, Matplotlib, Seaborn
- Web Development: Flask, HTML, CSS, JavaScript
- Tools & Platforms: Git, GitHub, Linux, Docker
- Cloud & Databases: AWS (Basic), SQL, QuestDB

PROJECTS

- **MAITRI — Multimodal AI Assistant for Astronauts:** Developed an AI-powered multimodal assistant to provide real-time psychological and emotional support for astronauts during space missions. Integrated CNN-based facial emotion recognition, CNN-LSTM speech emotion recognition, and Transformer-based text emotion analysis to detect emotions from face, voice, and text inputs. Built a multimodal fusion engine for stress analysis and adaptive conversational responses using TinyLlama. Implemented a critical emotion alert system with cooldown-based notifications, text-to-speech (TTS), and emotion logging for continuous monitoring. Developed the backend using Python, Flask, PyTorch, Hugging Face Transformers, OpenCV, Librosa, and Scikit-learn, enabling real-time inference and a ground control dashboard for alert monitoring. Designed the system for applications in space missions, healthcare, defense, and remote mental health monitoring.
- **Credit Card Fraud Detection System:** Developed machine learning and deep learning models, including **CNN, Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN)**, to detect fraudulent transactions in highly imbalanced financial datasets. Applied data preprocessing, feature engineering, class weighting, and threshold optimization to improve detection performance, and deployed the solution as a **Flask-based web application** for real-time fraud risk prediction.
- **House Price Prediction System:** Built a Linear Regression-based machine learning model to predict residential property prices using structured real estate data. Performed data preprocessing, feature selection, and model evaluation, and developed a web-based application that enables users to estimate house prices through real-time predictions.
- **Customer Segmentation System:** Developed a customer segmentation solution using the K-Means clustering algorithm to analyze purchasing behavior and identify distinct customer groups. Performed data preprocessing, exploratory data analysis, and feature scaling to generate actionable business insights for targeted marketing and customer relationship management.

CERTIFICATIONS

- Credit Linked Program in AI & ML – Daksh Gurukul, IIT Guwahati, Apr 2025– Aug 2026
- Computer Vision – Simplilearn | Skillup, Nov 2025
- Natural Language Processing – Simplilearn | Skillup, Nov 2025
- Deep Learning – Simplilearn | Skillup, Apr 2025
- Python Essentials – Cisco, Apr 2025
- Machine Learning with Python – Coursera (IBM), Mar 2025